

3

Lock Map

Save

LOAD

Lock map to start working

1

Paderborn University



No resul...

2

OpenStreetM

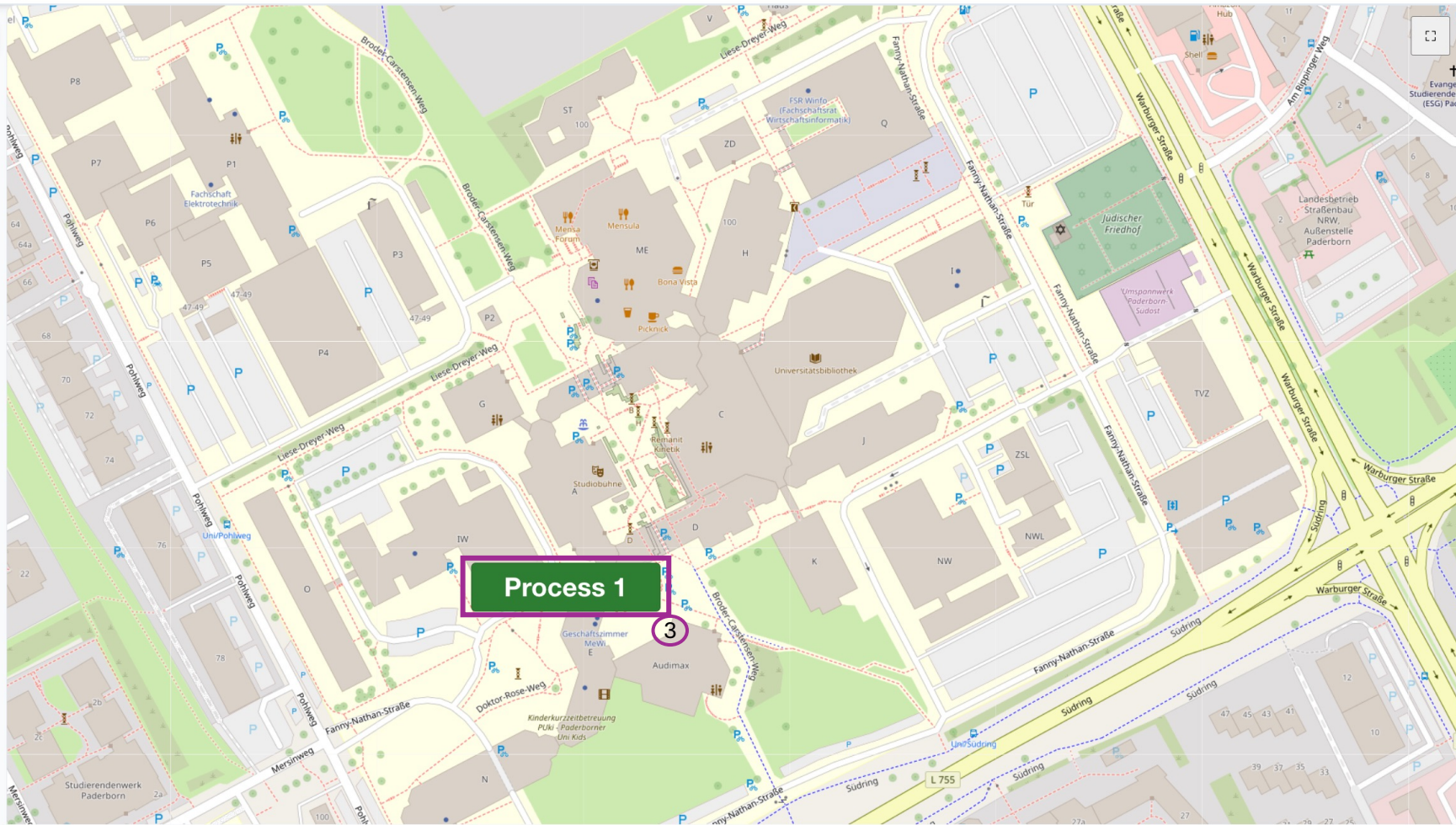
4



Placed successfully

OpenStreetMap

► **Process 1** **Place** 1 sub X



Unlock Map

Add Process

Save

LOAD

Measure

Export CSV

Locked Map

OpenStreetM

Process 1

Place

1 sub

Information

Subprocesses (1)

Show on Map

Add

Subprocess 1

Place

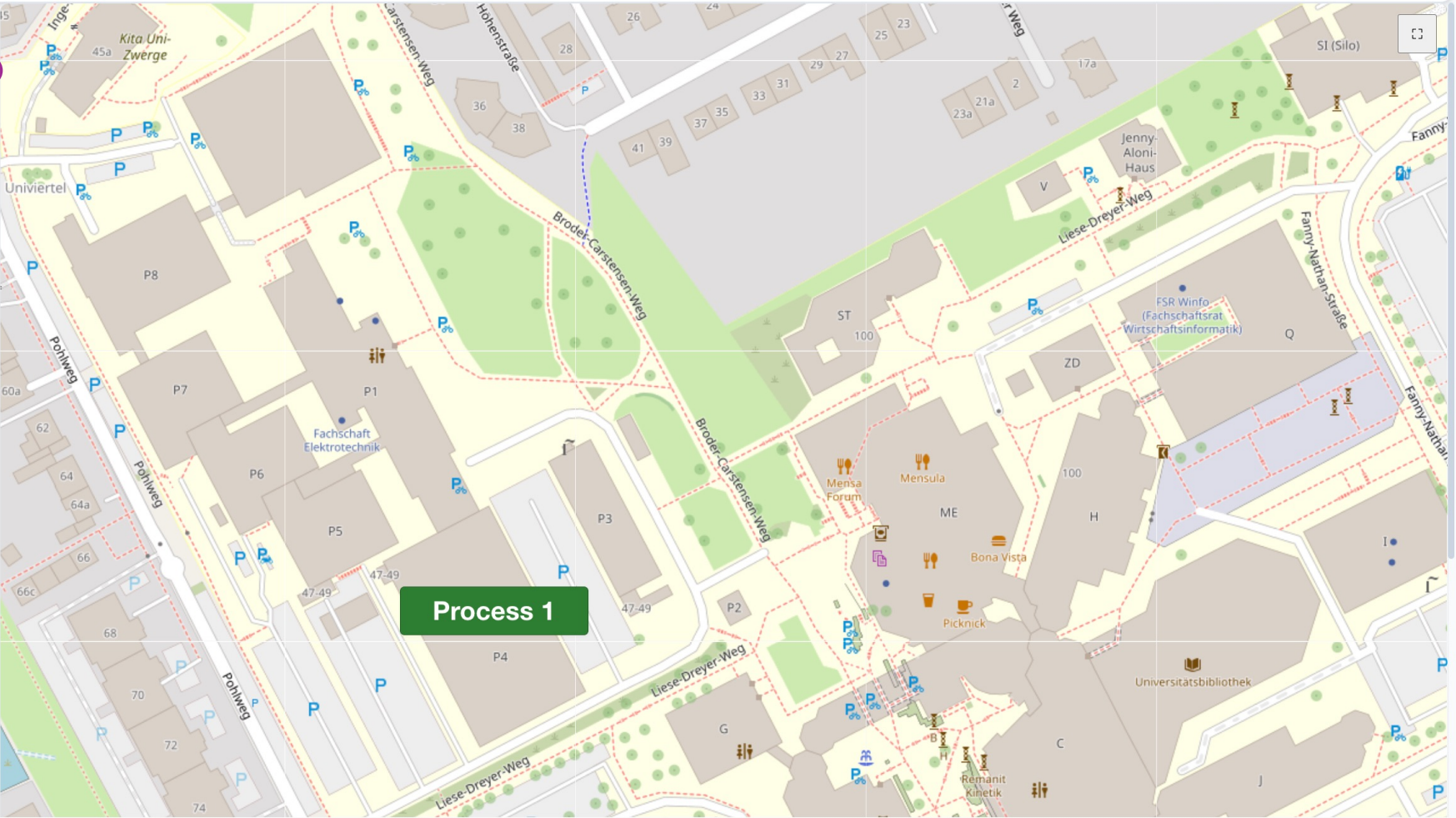
Next Processes (Click to link)

Streams

Notes

STREAMS (0)

Add Stream



[Unlock Map](#)[Add Process](#)[Save](#)[LOAD](#)[Measure](#)[Export CSV](#)

Locked Map

OpenStreetMap

Process 1

Place 4 sub

Information

Subprocesses (4)

Hide Map

Add

Subprocess 1

1

Next Processes (Click to link)

Streams

Notes

STREAMS (2)

2

Add Stream

Steam Subprocess 1 s2

Steam

Tin Tout m cp CP Water Content In

Water Content Out Density Pressure

TIN °C M kg/s

103 20

Product Subprocess 1 s1

Product

Tin Tout m cp CP Water Content In

Water Content Out Density Pressure

TIN °C TOUT °C M kg/s

43 23 5

CP kJ/(kg·K)

3

Subprocess 2

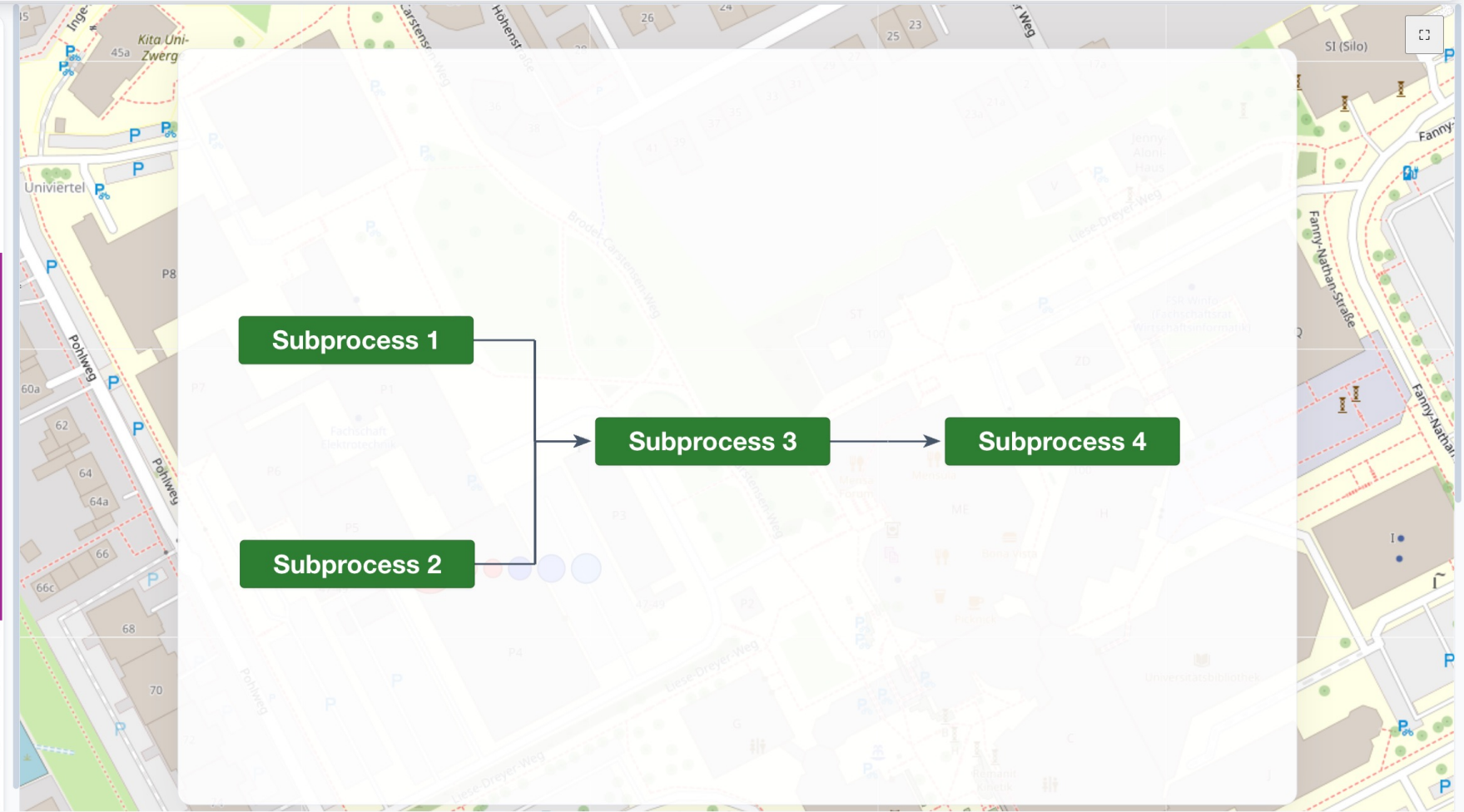
Place

Subprocess 3

Place

Subprocess 4

Place



Potential Analysis

All streams

Scenario

Save

Save

LOAD

Report



Streams Selection



Subprocess 1

- ☒ STEAM SUBPROCESS 1 S2 TIN:103°C | TOUT:43°C | CP:40.00 | Q:2400.00 KW |
- ☒ PRODUCT SUBPROCESS 1 S1 TIN:43°C | TOUT:23°C | CP:15.00 | Q:300.00 KW |

Subprocess 2

- ☒ PRODUCT SUBPROCESS 4 S3 TIN:128°C | TOUT:103°C | CP:15.00 | Q:375.00 KW |

Subprocess 3

- ☒ UTILITY SUBPROCESS 5 S4 TIN:88°C | TOUT:113°C | CP:30.00 | Q:750.00 KW |
- ☒ PRODUCT SUBPROCESS 1 TIN:68°C | TOUT:88°C | CP:50.00 | Q:1000.00 KW |

Subprocess 4

- ☒ SUBPROCESS 6 PRODUCT 6 TIN:8°C | TOUT:68°C | CP:20.00 | Q:1200.00 KW |

Current energy supply



Supply 1

HEAT (KW) 3000

Cold streams selection

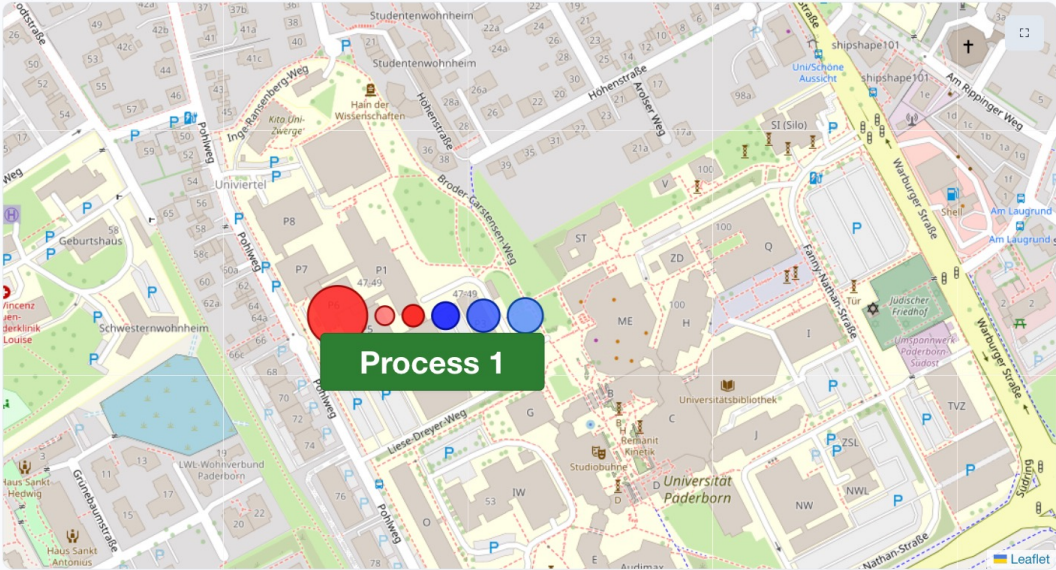
- ☒ UTILITY SUBPROCESS 5 S4
- ☒ PRODUCT SUBPROCESS 1
- ☒ SUBPROCESS 6 PRODUCT 6

COOLING (KW) 3000

Hot streams selection

- ☒ STEAM SUBPROCESS 1 S2
- ☒ PRODUCT SUBPROCESS 1 S1
- ☒ PRODUCT SUBPROCESS 4 S3

+ Add Supply



Comparison of Status Quo and Proposal



TYPE OF ENERGY DEMAND	CURRENT STATUS QUO (KW)	MIN. DEMANDS AFTER HEAT RECOVERY (KW)	SAVINGS (KW)	SAVINGS (%)	HP COVERAGE (KW) – VHTHP (HFC/HFO)	HEAT PUMP COVERAGE (%)
Heating Demand	3000.0	375.0	2625.0	87.5%	375.0	100.0%
Cooling Demand	3000.0	500.0	2500.0	83.3%	257.0	51.4%

SHOW SHIFTED COMPOSITE CURVES

ΔTMIN

10



Comparison of Status Quo and Proposal

1

TYPE OF ENERGY DEMAND	CURRENT STATUS QUO (KW)	MIN. DEMANDS AFTER HEAT RECOVERY (KW)	SAVINGS (KW)	SAVINGS (%)	HP COVERAGE (KW) – VHTHP (HFC/HFO)	HEAT PUMP COVERAGE (%)
Heating Demand	3000.0	375.0	2625.0	87.5%	375.0	100.0%
Cooling Demand	3000.0	500.0	2500.0	83.3%	257.0	51.4%

☐ SHOW SHIFTED COMPOSITE CURVES

ΔT_{MIN} 10

MINIMUM HEATING DEMAND

375.00 kW

MINIMUM COOLING DEMAND

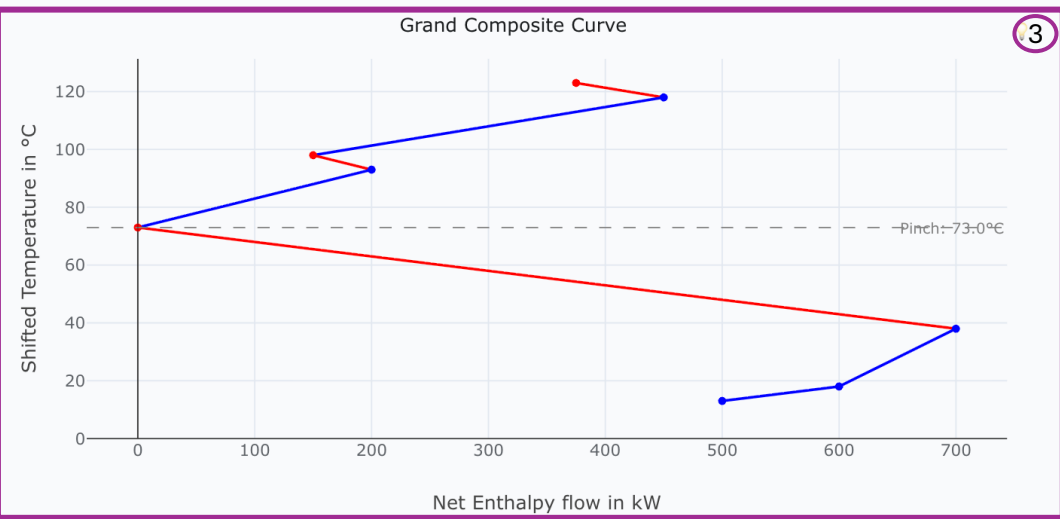
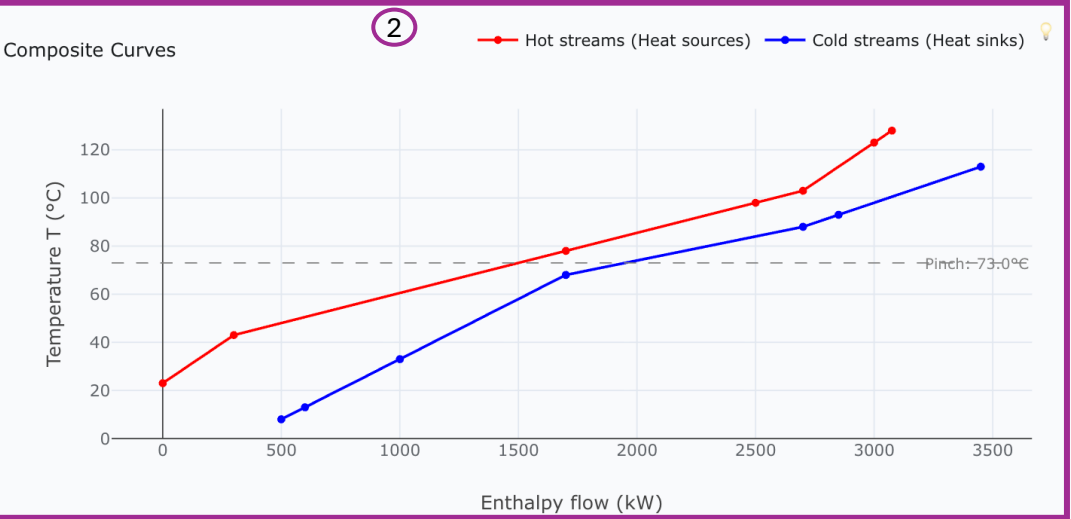
500.00 kW

PINCH TEMPERATURE

73.0 °C

HEAT RECOVERY POTENTIAL

2700.00 kW



INTEGRATION WITH HEAT PUMP MODEL AND OPTIMIZATION

REFRIGERANT TYPE	MEDIUM SINK	COP	Q_SINK (KW)	Q_SOURCE (KW)	ACTUAL SINK TEMP [°C]	SHIFTED SINK (T+5K) [°C]	ACTUAL SOURCE TEMP [°C]	SHIFTED SOURCE (T-5K) [°C]	REFRIGERANT	HP STAGES
Natural	Water	4.34	375.0	288.6	113.0	118.0	58.6	53.6	Butane	2
Synthetic	Water	4.18	375.0	285.3	113.0	118.0	58.7	53.7	R1224yd	2

Click on any column header to sort. Click on any point in the chart to inspect its details and add it to the table.

MEDIUM SINK

All

REFRIGERANT TYPE

All

REFRIGERANT

All

HP STAGES

All

Select All

Clear Selection

☐ PLOT SOURCE-LIMITED POINTS (7)

THESE COVER ONLY PART OF THE DEMAND ABOVE T_SINK, SO THEY LIE OFF THE SINK PROFILE.

☐ PLOT DEMAND-LIMITED POINTS (9)

THESE COVER THE FULL DEMAND BUT LEAVE WASTE HEAT UNUSED AT T_SOURCE, SO THEY LIE OFF THE SOURCE PROFILE.

Feasible Combinations: 1635
shown in chart: 1619

COP SOURCE

f Use my own COP formula

Paste a custom COP to perform integration and optimization with your own heat pump.

